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SafeStep

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Planning Stage

1.Project Idea and Justification

SafeStep is a smart mobile application designed to improve personal safety for women by providing fast and reliable emergency assistance features. The application allows users to quickly send alerts to trusted contacts using a one-touch emergency button. Once activated, the app automatically shares the user's live GPS location, sends emergency notifications, and begins audio recording to document the situation.

The main goal of SafeStep is to help women feel safer when they are alone, traveling, or in potentially dangerous situations. In many cases, women may not have enough time to make a phone call during an emergency. SafeStep simplifies the process by allowing them to request help instantly with just one tap.

Personal safety is an important issue in many communities. Many existing emergency solutions are either difficult to access quickly or not specifically designed to address women's safety concerns. SafeStep aims to provide a simple, fast, and user-friendly solution that enables immediate communication and location sharing during emergencies.

By combining modern mobile technology with a clear focus on safety, SafeStep contributes to building smarter and safer communities while supporting digital innovation in personal security.

2.Problem Statement

Despite the advancement of mobile technology, personal safety for women remains a significant concern in many urban and isolated environments. The core problem lies in the inefficiency of traditional emergency response methods during high-stress situations.

Conventional emergency calls often require multiple steps—unlocking the phone, dialing a number, and verbally explaining the location—which may be impossible if the user is being threatened or is in shock.

Furthermore, many existing safety apps are either too complex for rapid use or lack integrated features like automatic evidence collection (audio recording) and real-time location broadcasting. This technological gap leaves women vulnerable during critical moments where every second counts. There is a clear need for a specialized system that bypasses these barriers to provide an instantaneous, discreet, and reliable lifeline to trusted contacts and authorities.

3.Scope of Work

The SafeStep application focuses on providing emergency assistance features for women through a simple and reliable mobile platform. The system allows users to register their personal information and add trusted emergency contacts.

When the emergency button is activated, the application automatically captures the user's current GPS location, starts audio recording, and sends an alert notification to the selected contacts. The contacts will receive the user's location and emergency message, enabling them to respond quickly.

The system aims to provide fast communication during emergency situations while maintaining a simple and accessible user experience.

4.project time line:

Week 3-5
Planning and Requirement Analysis

Documenting the system requirements and analysing the problem related to women's personal safety.

Week 6-7
System Structure

Designing the system architecture and planning the main components such as the emergency button, GPS location sharing, and notification system.

Week 8-10
Application Development

Developing the main modules of the application including user registration, adding trusted contacts, and activating the emergency alert system.

Week 11-15
System Review

Testing the application functions, verifying system performance, and preparing the final project documentation.

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Analysis Stage

1.Introduction:

SafeStep is a smart mobile application designed to enhance women's safety during emergency situations by providing quick and reliable ways to request help. It allows users to send instant alerts and share their live location with trusted contacts to ensure a fast response.

2.Functional Requirements:

1. User Registration and Login:

Action:	The application allows the user to create a new account or log in securely using their personal credentials to access the system features.
Requirements:	User credentials must be stored securely and authentication must be validated before access is granted.
Pre-condition:	The user has installed the application.
Post-condition:	The user successfully logs into their account and can access all features.
Side effects:	None.

2.Manage Emergency Contacts:

Action:	The application allows the user to add, edit, and delete trusted emergency contacts who will receive alerts during emergency situations.
Requirements:	Contact information must be stored correctly and updated in the system database.
Pre-condition:	The user is logged into the application.
Post-condition:	Emergency contacts are saved and ready to receive alerts.
Side effects:	None.

3. Activate Emergency Alert:

Action:	The application enables the user to activate an emergency alert instantly using a single button to notify selected contacts.
Requirements:	The system must send alerts immediately and reliably without delays.
Pre-condition:	Emergency contacts are already added.
Post-condition:	Alert notifications are sent to all selected contacts.
Side effects:	Audio recording may start automatically.

4. Share Live Location:

Action:	The application allows the user to share their real-time GPS location with emergency contacts during an alert.
Requirements:	Accurate GPS tracking must be enabled.
Pre-condition:	Location services are enabled on the device.
Post-condition:	Contacts receive the user's live location.
Side effects:	Increased battery usage.

5. Audio Recording:

Action:	The application automatically records audio during emergency situations to provide evidence and additional context.
Requirements:	The system must store audio securely.
Pre-condition:	Emergency alert is activated.
Post-condition:	Audio recording is saved.
Side effects:	Consumes device storage.

6. Cancel Alert:

Action:	The application allows the user to cancel an emergency alert in case it was triggered accidentally.
Requirements:	The system must stop notifications immediately.
Pre-condition:	An alert is active.
Post-condition:	Alert is cancelled and contacts are notified.
Side effects:	None.

7. View Alert History:

Action:	The application allows users to view a history of previous alerts and related details.
Requirements:	The system must store alert logs.
Pre-condition:	User is logged in.
Post-condition:	User can access alert history.
Side effects:	Audio recording may start automatically.

3. Non-Functional Requirements:

1. Product Requirement:	The system shall ensure data security and privacy by protecting user information such as location and emergency contacts using secure storage and encryption techniques.
2. Performance Requirement:	The system shall provide fast response times by sending emergency alerts instantly and sharing the user's live location without noticeable delay.
3. Usability Requirement:	The system shall be easy to use by providing a simple and clear user interface that allows users to activate emergency alerts quickly, even in stressful situations.
4. Reliability Requirement:	The system shall be reliable and stable by ensuring that emergency alerts are delivered successfully without system crashes or failures.
5. Availability Requirement	The system shall be available 24/7 to ensure that users can access emergency features at any time without interruptions.
6. Accuracy Requirement:	The system shall provide accurate GPS location tracking to ensure that emergency contacts receive the correct real-time location of the user.
7. Compatibility Requirement:	The system shall be compatible with different mobile devices and operating systems (Android and iOS) to ensure that all users can access the application without technical issues.

4.User Personas:

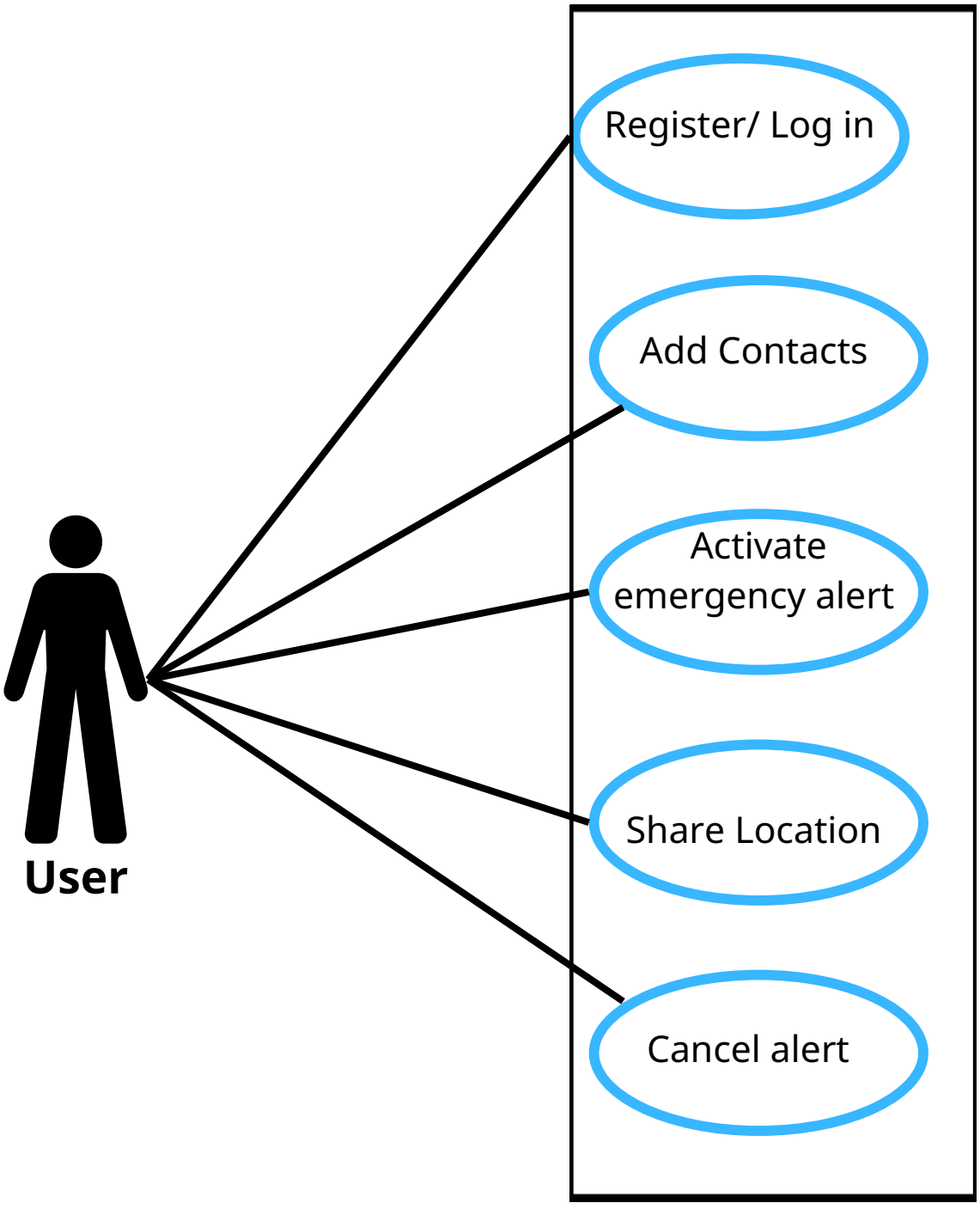
Employee Persona:
Name: Layan
Age: 25
Occupation: Marketing Specialist
Needs: She needs quick help because she uses different transportation during her daily commute and may feel unsafe between locations.

Student Persona:
Name: Raghad
Age: 20
Occupation: University Student
Needs: She needs quick help because she leaves the university at night and may face unsafe situations.

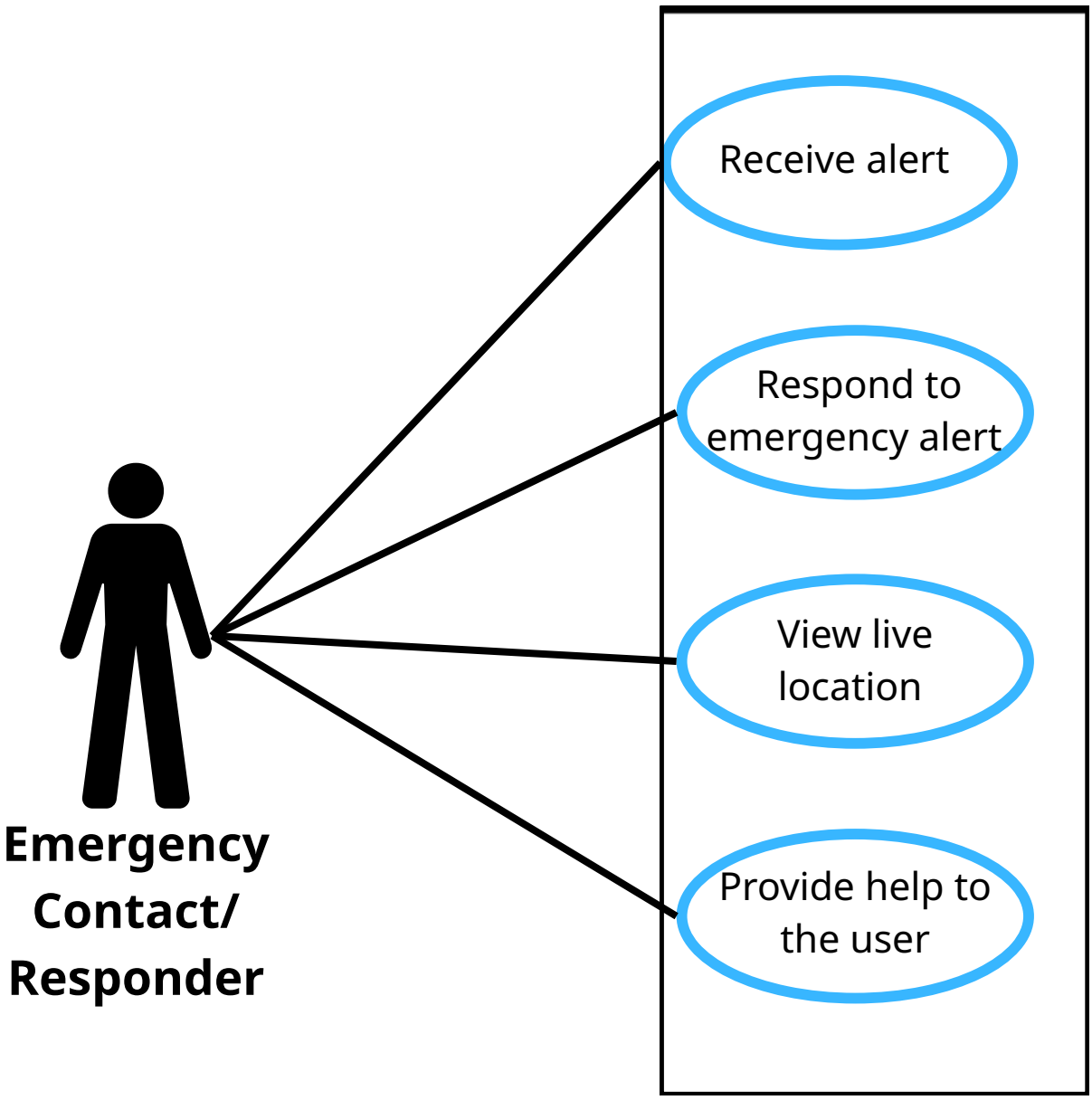
Nurse Persona:
Name: Jumana
Age: 28
Occupation: Nurse
Needs: She needs quick help because she works late shifts and may feel unsafe when returning home at night.

University Professor Persona:
Name: Foziah
Age: 32
Occupation: Assistant Professor
Needs: She needs quick help because she works long hours at the university and may feel unsafe when leaving campus late.

5.Use Case Diagram:



5.Use Case Diagram:



6. Risks and Mitigations:

Identifying potential risks is crucial to ensure the reliability of SafeStep and to maintain the trust of users during critical moments

Risk Category	Risk Description	Mitigation Strategy
Data Privacy	Potential leakage of sensitive location data or contact information	Implement End-to-End Encryption and strictly adhere to data protection standards to ensure (like GDPR) only authorized contacts can access the location
System Availability	Server downtime or app crashes when a user attempts to send an alert	Utilize High-Availability Cloud Services (99.9% uptime) and implement offline capabilities for basic alert triggering
GPS Accuracy	Inaccurate location tracking in indoor environments or tunnels	Integrate multi-source -positioning, such as Wi Fi Positioning and Cell Tower Triangulation, to supplement GPS data

6. Risks and Mitigations:

Identifying potential risks is crucial to ensure the reliability of SafeStep and to maintain the trust of users during critical moments

Risk Category	Risk Description	Mitigation Strategy
False Alarms	Frequent accidental alerts, which may lead emergency contacts to ignore real emergencies	Include a Short ,.Delay/Countdown (e.g seconds) before the 5 alert is sent, allowing the user to cancel accidental triggers via biometric authentication
Battery Consumption	High battery drain due to continuous background location tracking	Optimize the app to remain in "Low Power Mode" and only activate high-precision tracking when a trip starts or an alert is triggered
Response Reliability	Emergency contacts failing to notice the alert e.g., if their phone is on) (silent	Use Critical Alerts features (available on iOS/Android) that bypass silent mode to ensure the notification is heard by the responder

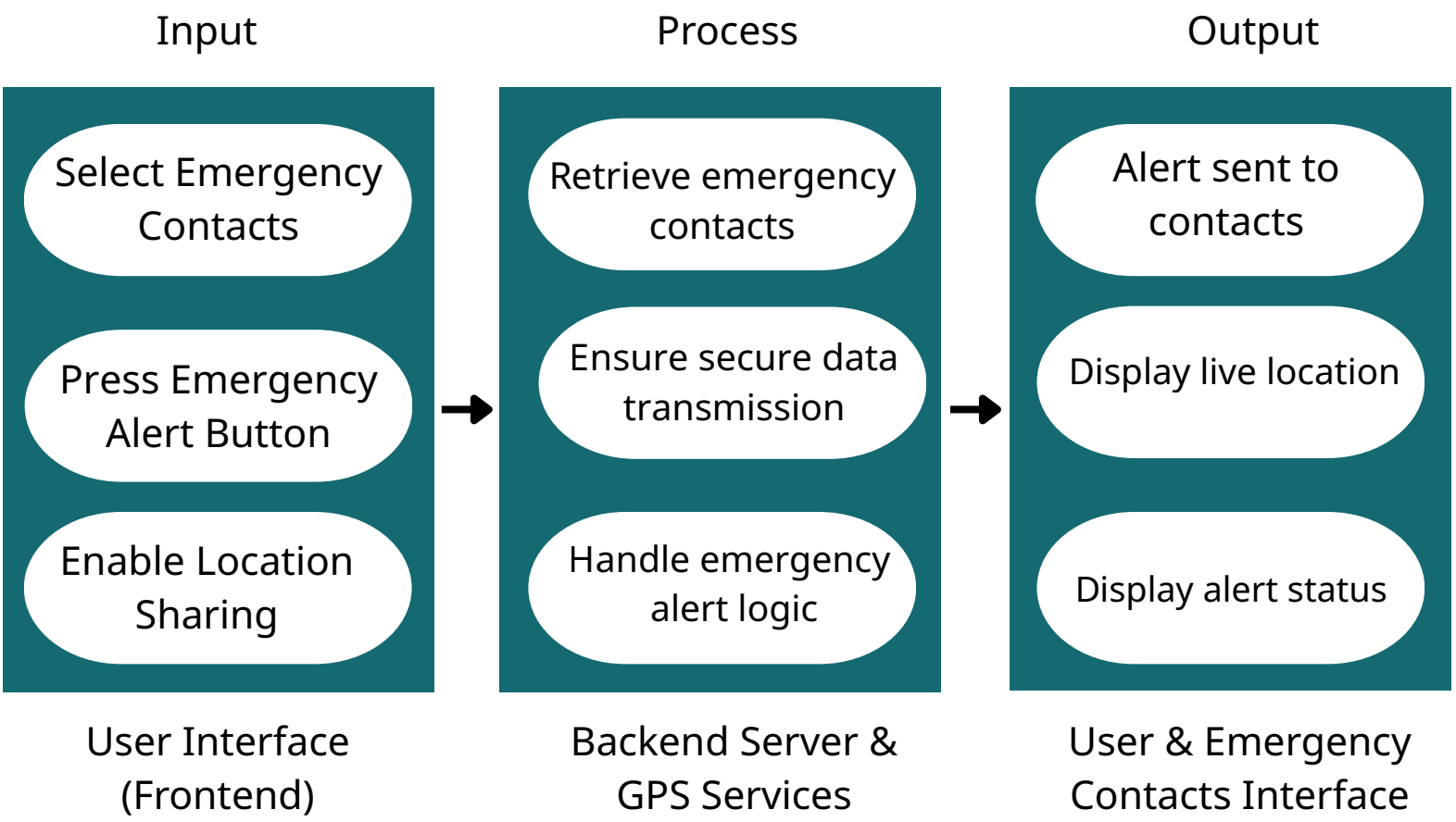
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Design Stage

1.Introduction:

SafeStep moves from the analysis phase to the design stage, where system ideas are transformed into a structured solution. This stage focuses on designing a clear architecture and user-friendly interfaces to ensure a reliable and easy-to-use safety application.

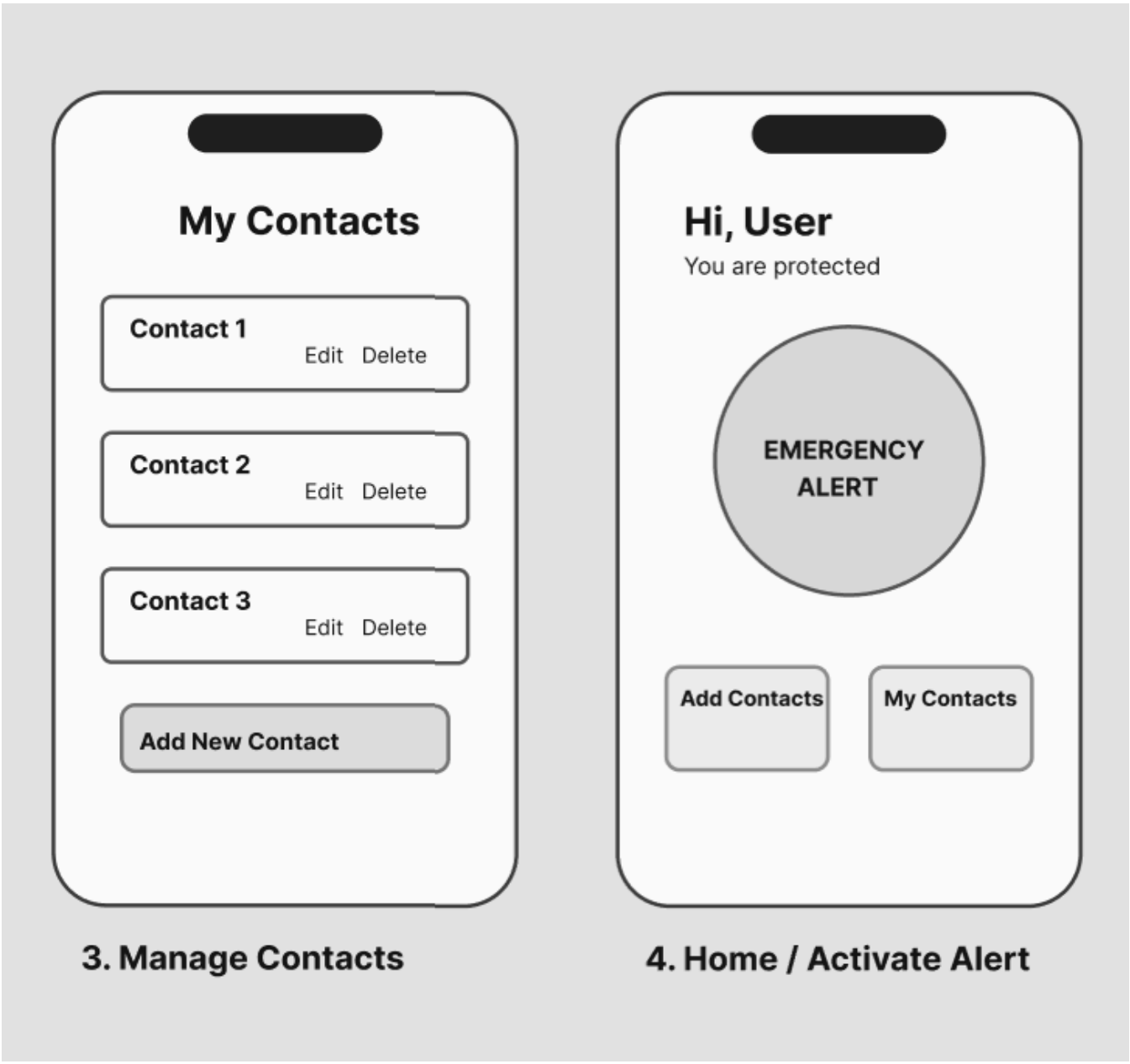
2.Software Architecture Overview:



3.UI Mockups:


SafeStep
Email or Phone
Password
Login
Create New Account
1. Register / Login
Add Contact
Contact Name
Phone Number
Relationship
Save Contact
2. Add Emergency Contacts

3.UI Mockups:


My Contacts
Contact 1
Edit Delete
Contact 2
Edit Delete
Contact 3
Edit Delete
Add New Contact
Hi, User
You are protected
EMERGENCY ALERT
Add Contacts
My Contacts
3. Manage Contacts
4. Home / Activate Alert

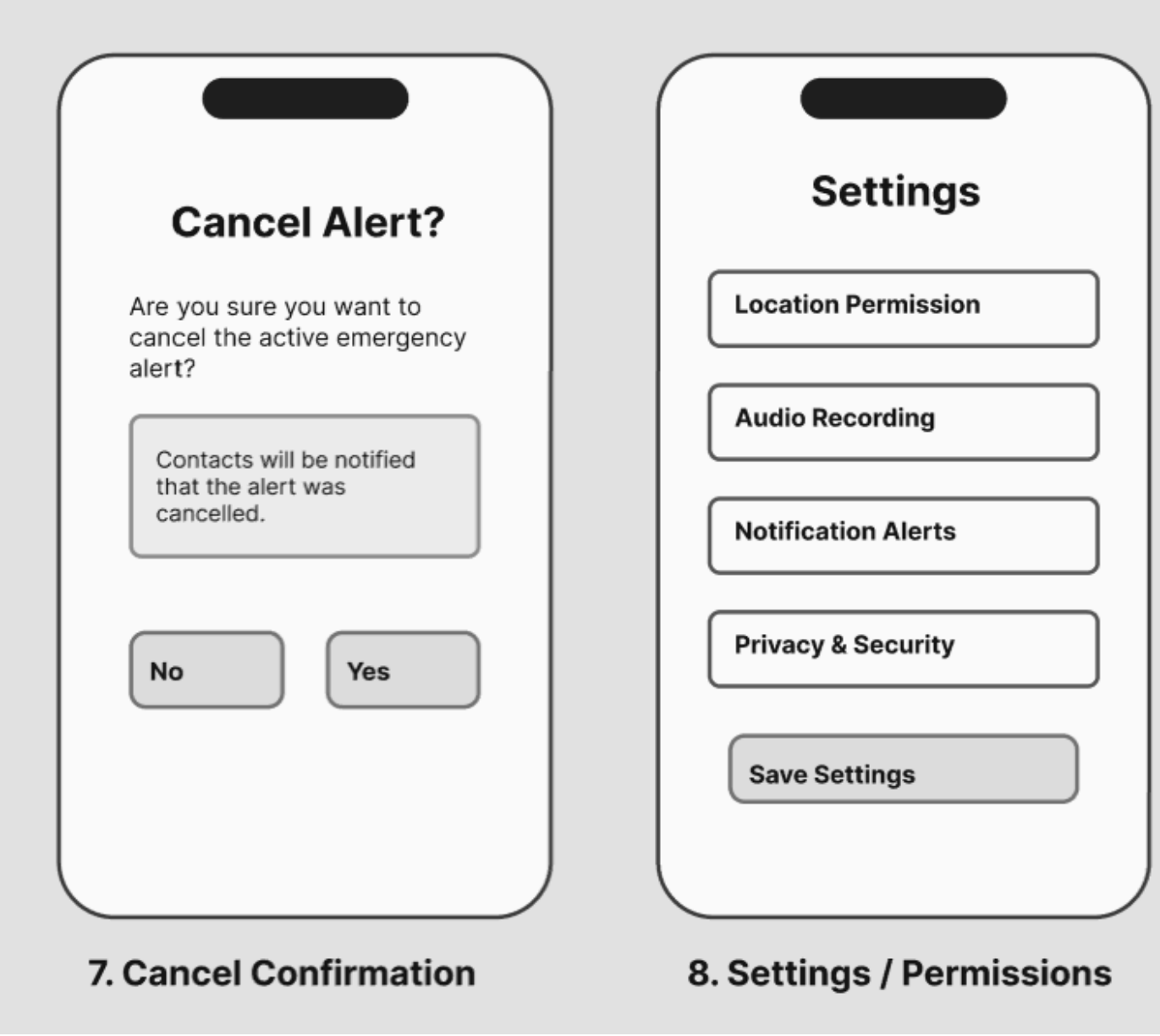
3.UI Mockups:



5. Emergency Active

6. Alert History

3.UI Mockups:



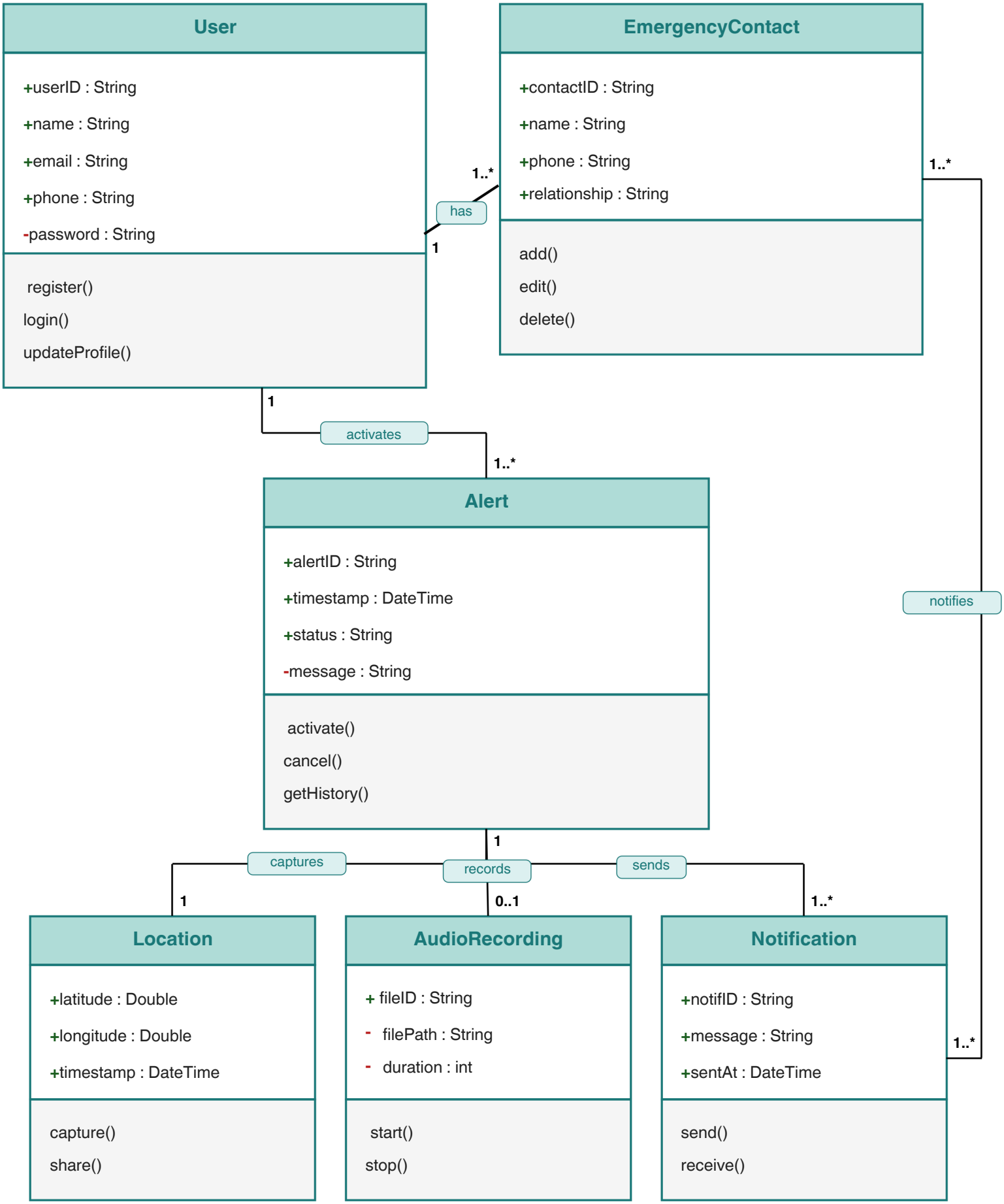
7. Cancel Confirmation

8. Settings / Permissions

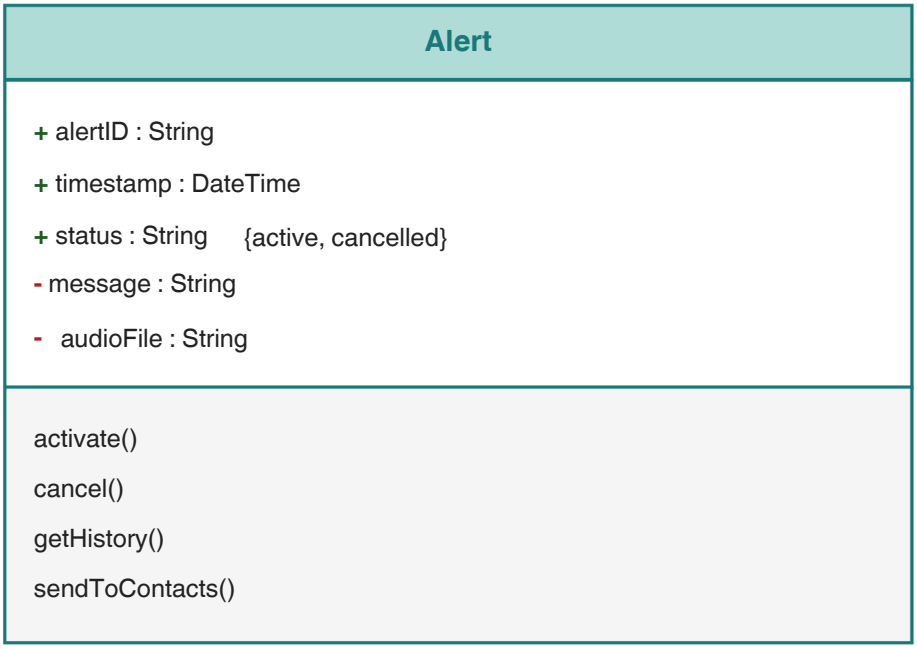
4. class diagram:



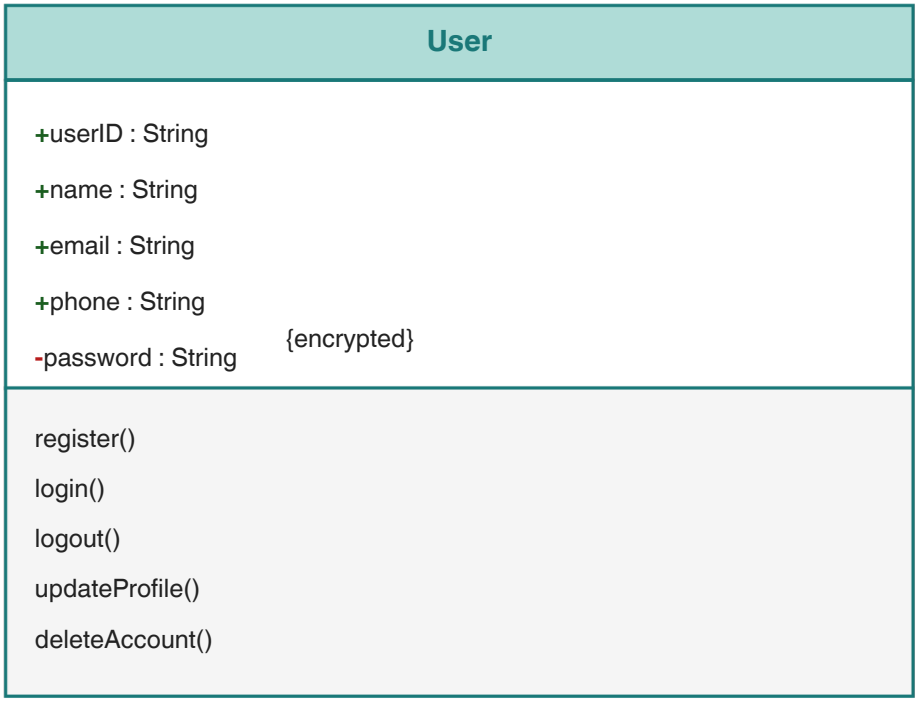
Classes and Associations in the SafeStep App



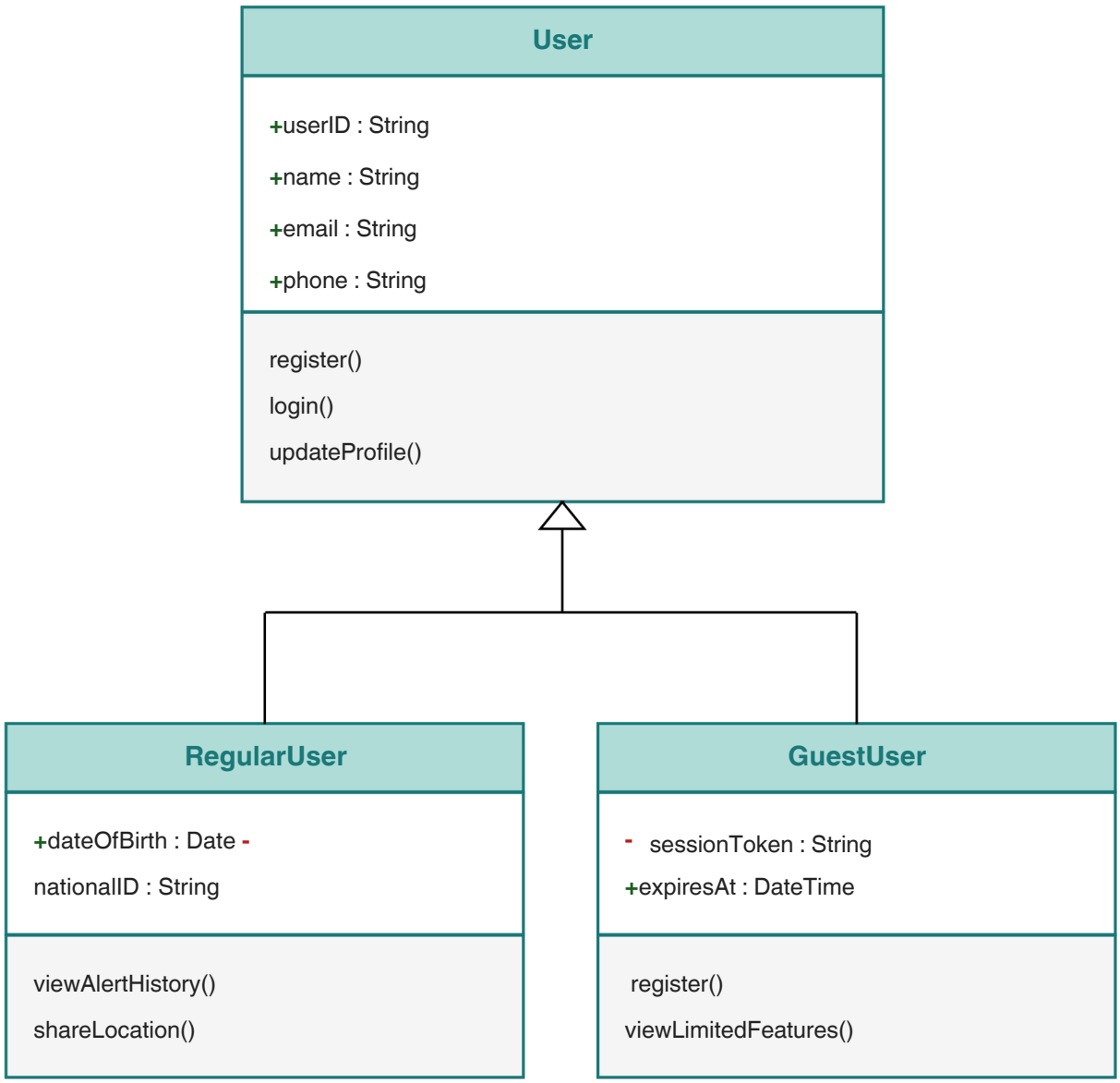
The Alert Class



The User Class



A Generalization Hierarchy with Added Detail





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Testing Stage

1. Test Scenarios:

TS01: Create Account Successfully
The system should successfully distinguish between a new account and an existing account and allow registration only for new users.

TS02: Emergency Alert Notification Sent Successfully
The system should successfully send emergency alert notifications to the selected emergency contacts.

TS03: Live Location Sharing Failure
The system may fail to retrieve or share the user's live location on some devices due to GPS or device compatibility issues.

TS04: Missing Emergency Contact Information Validation Failure
The system fails to prevent the user from saving emergency contacts with incomplete required information.

2. Test Cases :

Test Case 1 :

Test Case ID: TC01
Related Scenario: TS01

Description: Verify that the system allows a new user to create an account successfully.

Steps:
1. Open the SafeStep application.
2. Click on “Register”.
3. Enter a new email address.
4. Enter a valid password.
5. Confirm the password.
6. Click the “Create Account” button.

Expected Result:
The system should create the account successfully and redirect the user to the home page.

Actual Result: Pass

2. Test Cases :

Test Case 2 :

Test Case ID: TC02
Related Scenario: TS02

Description: Verify that the system sends emergency alert notifications to selected contacts.

Steps:
1. Open the SafeStep application.
2. Log in to the system.
3. Add at least one emergency contact.
4. Press the emergency alert button.
5. Allow location access if requested.

Expected Result:
The selected emergency contacts should receive an emergency alert notification with the user’s live location.

Actual Result: Pass

2. Test Cases :

Test Case 3 :

Test Case ID: TC03
Related Scenario: TS03

Description: Verify the system behavior when live location sharing fails.

Steps:
1. Open the SafeStep application.
2. Disable GPS or location services on the device.
3. Press the emergency alert button.
4. Attempt to share the live location.

Expected Result:
The system should display an error message indicating that location sharing is unavailable.

Actual Result : Pass

2. Test Cases :

Test Case 4 :

Test Case ID: TC04
Related Scenario: TS04

Description: Verify that the system validates incomplete emergency contact information.

Steps:
1. Open the SafeStep application.
2. Navigate to the “Emergency Contacts” section.
3. Click “Add Contact”.
4. Leave one required field empty (such as phone number).
5. Click the “Save” button.

Expected Result:
The system should display a validation message and prevent saving incomplete contact information.

Actual Result: Fail

3.Bug Report:

Bug ID: B01

Title: Emergency Alert Notification Not Sent

Description:
When the user presses the emergency alert button, the system sometimes fails to send the alert notification to emergency contacts.

Steps to Reproduce:

1. Open the SafeStep application.
2. Log in using a valid account.
3. Add an emergency contact.
4. Press the emergency alert button.
5. Wait for the notification process.

Expected Result:
The system should send the emergency alert notification and live location to the selected emergency contacts.

Actual Result:
No emergency alert notification is received by the emergency contacts.

Severity: Medium

Status: Pass

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